HINA SUZUKI

hinas@arizona.edu ◇ GitHub

EDUCATION

The University of Arizona Tucson, AZ
Bachelor of Science in Astronomy with minor in Electrical and Computer Engineering December 2025
Major sGPA: 3.88/4.00

Relevant Coursework:

Astronomy:	Theoretical Astrophysics I (Stars) & II (Galaxies & Cosmology), Observational Astronomy, Astrophysics II (Radiation & Matter)
Physics:	Quantum Mechanics, Electromagnetism, *General Relativity, Computational Physics, Thermal Physics
Engineering:	Computer Programming I&II, Computer Aided Design, Digital Logic, *Quantum Computing and Quantum Information, Deep Learning, *Quantum Information Processing and Quantum Error Correction

* are graduate level courses.

TECHNICAL SKILLS

Programming Languages:	C/C++; Python; MATLAB;
Software:	Git, LaTeX; JAX, High Performance Computing, PyTorch, Qiskit, Bash, SolidWorks, KiCad.
Instruments:	Spectrum analyzer; Analog to digital converter; Oscilloscope; Multimeter;

RESEARCH

Event Horizon Telescope Data & Science
Steward Observatory, The University of Arizona January 2023 - Present

Project: **High-Order Photon Ring Imaging of Naked Singularity**
Advisor: Dr. Chi-Kwan Chan, Dr. Yosuke Mizuno
Goal: To study the High-Order photon ring of the Naked Singularity under variable observations, allowing analysis on shadow geometries. To run General Relativistic Radiative Transfer(GRRT) code for GRMHD simulated data of Naked Singularity on High Performance Computer, which allows higher resolution images and larger number of snapshots.
Results: Developed pipeline to analyze the photon ring geometry around the Kerr naked singularity. Ran GRRT calculation on High Performance Computer (CPU Hours:). We found that cascaded structure of photon rings form as predicted ones for the black holes, but uniquely forming rings inside the nominal shadow region. This result will be published in my first author paper in preparation.

Project: **Improved and refined the algorithm to calibrate fast data acquisition systems**
Advisor: Dr. Chi-Kwan Chan
Goal: To perform faster data acquisition with developing new system without tailored components. To improve algorithm to calibrate mismatches interleaving Analog-To-Digital Converter system. Though Event Horizon Telescope motivated this project, this algorithm has broad applications.
Results: Refined mathematical equation by applying Fourier transform. Performed numerical simulations to test the accuracy of calibration mismatches with and without any possible noise from the system. The paper of this work is in preparation.

Steward Observatory Radio Astronomy Laboratory
Steward Observatory, The University of Arizona January 2024 - Present

Project: **2.7 m Inflatable Aperture Telescope Mechanical Design and Fabrication**
Advisor: Dr. Christopher Walker
Goal: To design 2.7m telescope support structure to perform vacuum tolerance test of the telescope in a chamber. To design and review Radio Frequency transmission circuits in the detector to minimize the limit of observation due to electrical circuit.

EMPLOYMENT

National Radio Astronomy Observatory
Electronics Department, National Radio Astronomy Observatory — May 2024 - August 2024
Supervisor: David Schafer

Project: **Motor Control System Rehabilitation and GUI Development**
Project Link for Motor Control GUI: https://github.com/hina18201716/RF-DFS/tree/gui-dev
Rehabilitated a motor control system electronics to control the 3-m dish antenna located at the Very Large Array (VLA) site, and refined electrical wiring with formatted wiring diagram. Developed Graphical User Interface(GUI) to control the antenna online. Developed foundation of interface between Direction-Finding antenna and Environmental Monitoring System to search Radio interference sources more effectively. Confirmed motion of the antenna through command line program with the built-in package.

Project: **Upgrade of Visitor Center Solar Telescope**
Project Link for Solar Telescope: https://github.com/hina18201716/Solar-Radio-Telescope
Upgraded the solar radio telescope by the visitor center on the site of the VLA. Developed an LED array for a new indication of the radio signal captured by the telescope. Designed comparator/Analog-to-Digital circuits with Op-amps to drive LED array. Assembled a PCB (Printed Circuit Board). Confirmed predicted behavior of the circuit. Designed the solar finder on 3D CAD. Organized team to fabricate and install the solar finder.

Engineering Technician
FreeFall Aerospace, Tucson, AZ — May 2025 - Present
Supervisor: Aman Chandra

Project: **All Sky Antenna Feed Motor Control Software Development**
–

TALKS, WORKSHOPS

247th AAS meeting (Abstract Submitted)
"High-order Photon Rings around Naked Singularities" — January 2026
Phoenix, Arizona

Steward Symposium *"Interleaving ADC"* — November 2024
Steward Observatory, The University of Arizona

PROFESSIONAL DEVELOPMENTS

Black Hole Explorer Science Meeting — September 2025
Virtual

AstroTech Summer School — June 2025
Department of Astronomy, University of California, Berkeley

Black Hole Explorer Japan Workshop — June 2024
Virtual (National Astronomical Observatory of Japan)

COURSE PROJECTS

MW-M31 Major Galaxy Merger Remnant Rotation Profile — Spring 2025
Advisor: Dr. Gurtina Besla
Project Link for the project: https://github.com/hina18201716/astr400b-25s/tree/main/ResearchAssignments

Quantum Kernel Estimation Literature Review — Spring 2025
Advisor: Dr. Narayanan
Project Link for QKE Class Project: https://github.com/hina18201716/quantum-machine-learning

HONORS & ACHIEVEMENTS

Department of Astronomy Outstanding Senior Award 2025
awarded to one top graduating senior for excellence in academics, research/internship involvement, and leadership within the campus or broader community.

Undergraduate Research Travel Grant 2025
awarded to up to 20 students across college, to support support undergraduate students in presenting or attending conferences and professional development events across disciplines.

Galileo Circle Scholarship 2025
awarded to the top 1.5% of undergraduate and graduate students in the College of Science, recognizing scholastic achievement, research excellence, and strong character.

Angelos C. Langadas Scholarship in Astronomy 2025
–

William F. Lucas Astronomy Scholarship 2024
–

Wildcat Tuition Award 2022
4-year tuition merit-based scholarship ($18,000 per year) for outstanding incoming international students

LEADERSHIP

Tucson Initiative for Minoritized student Engagement in Science and TEchnology Program Tucson, AZ
TIMESTEP Leader August 2024 - Present
Link for TIMESTEP: https://timestep.arizona.edu, dedicated to help minority students in astronomy and physics at University of Arizona.

- Presented academic experiences and strategies to get research experience, internships and scholarships to lead peers to success.
- Provided feedback of the writing applications for students applying research position, internships, and other opportunities.

LANGUAGES

English (Fluent); Japanese (Native);